

Resolving Coordination Frictions in Structural Labor Transitions

The Case of the Green Transition

Shisham Adhikari

University of California - Davis

ABSTRACT

This paper develops a unified framework to evaluate how policy can manage large-scale labor reallocations when coordination frictions between workers and firms slow adjustment. Extending the Diamond–Mortensen–Pissarides model, we introduce sector-specific entry costs for workers, production subsidies for firms, and positive externalities from green employment, and assess policies across four dimensions: their ability to meet reallocation targets, their effects on transitional unemployment, their fiscal cost, and their impact on aggregate welfare. Applied to the U.S. green transition, the analysis shows that worker-side or firm-side subsidies alone can raise green employment from 2% to 14% by 2030, but at the cost of higher unemployment and fiscal burdens. By contrast, a coordinated dual strategy aligns incentives on both sides of the market, lowering unemployment by up to 18% and reducing fiscal costs by more than 31% relative to one-sided approaches. These gains are further magnified by mild increasing returns in matching, while accounting for environmental externalities—equivalent to a 0.6% productivity boost—turns the transition into a net welfare improvement. The results demonstrate that treating workers and firms as co-dependent levers, rather than isolated targets, enables governments to design interventions that make structural shifts more efficient, fiscally sustainable, and socially inclusive.

This Version: October 2, 2025

1 Introduction

Managing large structural shifts, such as decarbonization, digitalization, and trade realignment, has become a central challenge for policymakers. These transitions create opportunities for long-term growth, but history shows they can also generate costly dislocations if left unmanaged. The adjustment to rising import competition from China in the United States, which left many regions with persistently high unemployment (Autor, Dorn, and Hanson, 2013), illustrates how sectoral reallocations can be disruptive. Similar risks arise today: the global commitment to carbon neutrality by 2050 requires moving labor out of fossil-fuel industries and into clean-energy sectors, a process unlikely to proceed smoothly without policy support. Poorly managed transitions risk straining fiscal resources, raising unemployment, and reducing aggregate welfare.

One of the central difficulties in any sectoral transition is the coordination problem between firms and workers. Workers face costs and risks related to training, relocation, and benefits when switching sectors, while firms hesitate to expand investment and jobs without assurance of a reliable labor force. This tension is already evident in green labor markets. Workers in carbon-intensive activities face rising uncertainty (Blue-dorn, Hansen, Noureldin, Shibata, and Tavares, 2023), while clean-energy employers report shortages of qualified labor. LinkedIn’s *Global Climate Talent Stocktake 2024* shows that postings for green jobs have grown nearly twice as fast as the available workforce (LinkedIn, 2024), and international organizations project multi-million worker shortfalls by 2030 (ILO, 2019; BCG, 2023). Although existing studies incorporate frictions, less attention has been given to this coordination failure—the interdependence of firms’ hiring decisions and workers’ transition choices—which can leave both sides waiting and stall reallocation.

This paper develops a policy framework to address labor reallocation when coordination frictions are present. The analysis is guided by four policy-relevant questions: (i) how can policy achieve targeted reallocation into new sectors, (ii) which policy mix minimizes transitional unemployment, (iii) which achieves these goals at the lowest fiscal cost, and (iv) which creates the fewest welfare distortions. To answer these questions, we extend a Diamond–Mortensen–Pissarises search-and-matching model with two sectors, where workers face entry costs to move into the green sector and firms receive production subsidies to expand vacancies.

Applied to the U.S. green transition, the framework tests three policies: firm-side support, worker-side support, and a dual strategy combining both. All three raise green employment from 2% to 14% by 2030 (WorkingNation, 2024), but their consequences dif-

fer sharply. One-sided policies achieve the employment target but at the cost of higher transitional unemployment and larger fiscal burdens. By contrast, the dual strategy lowers unemployment by up to 15 percent and reduces fiscal costs by more than 20 percent relative to one-sided measures. Two extensions strengthen this result. First, incorporating the positive externalities of green employment equivalent to at least a 0.6 percent increase in aggregate productivity shows that coordinated policies generate net welfare gains. Second, mild increasing returns to scale in the matching function amplify these efficiency gains, further shrinking unemployment, enhancing welfare, and lowering overall funding requirements.

The broader implication is that effective transition policy must align the decisions of firms and workers. By making the coordination problem the focus of analysis, this paper shows that dual-sided interventions achieve reallocation targets more effectively, while at the same time reducing unemployment, fiscal costs, and distortions. While the application here centers on the U.S. green transition, the framework offers general insights for managing structural shifts across economies.

2 Relevant Literature

Macroeconomic models of decarbonization have advanced understanding of how production shifts between carbon-intensive and green sectors, but they often assume frictionless labor mobility (Bilal and Stock, 2025). Evidence from Europe’s coal phase-outs and the slow reskilling in U.S. manufacturing shows instead that sectoral transitions are prolonged and costly. Early computable general equilibrium (CGE) models provide useful benchmarks for reallocation (Patuelli, Nijkamp, and Pels, 2005; Sancho, 2010; Böhringer, Rivers, and Rutherford, 2013; Freire-González, 2018), yet their reliance on full employment abstracts from unemployment, search costs, and mismatches. More recent approaches incorporate labor market frictions (Hafstead and Williams III, 2018; Hafstead, Williams III, and Chen, 2022), but still emphasize aggregate outcomes rather than the policy instruments that shape reallocation dynamics. Empirical studies reinforce these concerns. Walker (2011, 2013) document modest displacement under the Clean Air Act; Shapiro and Metcalf (2023) show that the long-run effects of carbon taxes depend on technology adoption; and Conte, Desmet, and Rossi-Hansberg (2022) highlight spatial heterogeneity in carbon pricing impacts. International evidence underscores displacement and distributional pressures during the green transition (ILO, 2019; Bluedorn et al., 2023).

A growing search-and-matching literature emphasizes hiring frictions and skill bot-

tlenecks (Gibson and Heutel, 2023; Lankhuizen, Rojas-Romagosa, and van Ewijk, 2022), but pays little attention to the coordination problem: workers delay retraining without credible job opportunities, while firms delay vacancies without a reliable labor pool. This dynamic—well established in macroeconomics (Cooper and John, 1988; Diamond, 1982)—remains underexplored in sectoral transitions. Policy studies on active labor market programs show that training, mobility support, and hiring subsidies can ease adjustment, but the comparative effectiveness of worker- versus firm-side interventions remains debated (Card, Kluve, and Weber, 2018; Brown, 2015; Kluve, Puerto, Robalino, Romero, Rother, Stöterau, Weidenkaff, and Witte, 2019), and these instruments are rarely embedded in general equilibrium analysis.

This paper bridges these literatures. We extend the search-and-matching tradition (Mortensen and Pissarides, 1994; Pissarides, 2000) to model the endogenous entry decisions of workers and firms, highlight their strategic interdependence, and evaluate three policies—firm subsidies, worker subsidies, and dual support—on reallocation speed, unemployment, fiscal cost, and welfare. We also provide the first analysis incorporating environmental externalities into a DMP framework, quantifying the productivity-equivalent threshold at which coordinated policies yield net welfare gains. By centering coordination failures and externalities, the paper shows why policy design is critical for efficient and fiscally sustainable labor reallocation in the low-carbon transition.

The remainder of the paper is organized as follows. Section 2 presents the model; Section 3 introduces policy instruments; Section 4 calibrates to U.S. data and reports results; Section 5 extends the analysis to increasing returns and externalities; Section 6 concludes with policy lessons and broader applications.

3 Model

We begin with a static benchmark of a modified Diamond–Mortensen–Pissarides (DMP) model to characterize equilibrium in the presence of coordination frictions that arise during green reallocation. The static environment allows a clear analysis of how subsidies and costs shape outcomes in segmented labor markets. Building on these insights, we then develop a dynamic version that characterizes equilibrium over time, allowing us to study unemployment dynamics, fiscal requirements, and aggregate welfare—central concerns for current green transition policies.

3.1 Static Model: DMP Framework

The static model provides a tractable setting to study how workers and firms interact across segmented labor markets when policy instruments target each side. We focus on (i) multiplicity arising from coordination frictions, (ii) how subsidies and costs affect equilibrium, and (iii) efficiency conditions.

3.1.1 Physical Environment

We consider a static (one-period) version of the DMP model (Pissarides, 2000) with two sectors: green (g) and non-green (n). The labor force is normalized to 1, and both workers and firms decide optimally which market to enter. A continuum of unemployed workers receive an unemployment benefit z . Let $\pi \in [0, 1]$ denote the share of unemployed workers who enter the green market (thus $1 - \pi$ choose the non-green market). Firms choose in which sector to post vacancies.

Firms that enter either sector incur a recruiting cost c . Green firms receive a production subsidy τ_g , while non-green firms face a tax τ_n that finances the subsidies.¹ On the worker side, entry into the green sector requires paying a flow cost κ_g , partially offset by a subsidy $s_g \in [0, 1]$. The effective cost for a worker who attempts to match in the green market is $(1 - s_g)\kappa_g$. Entry into the non-green sector is costless.

Matching within each sector $j \in \{g, n\}$ follows

$$f(u_j, v_j) = \delta_j \left(\frac{u_j v_j}{u_j + v_j} \right)^{1-\psi} (u_j v_j)^\psi, \quad \psi \in [0, 1], \quad (1)$$

where u_j and v_j are, respectively, unemployed workers and vacancies in sector j , and δ_j is sectoral matching efficiency. We assume constant returns to scale (CRS) with $\psi = 0$ in the baseline.² The implied matching probabilities are

$$\alpha_{wj} = \frac{v_j}{u_j + v_j}, \quad \alpha_{fj} = \frac{u_j}{u_j + v_j}, \quad j \in \{g, n\}. \quad (2)$$

After matches form, wages are set by Nash bargaining. Workers' bargaining power is denoted by $\eta \in (0, 1)$. All other primitives are identical across sectors.

This environment embeds the coordination problem salient in green labor markets.

¹We also solved a version in which all sectors are taxed to fund green subsidies. Since this does not change the qualitative results and adds notation, we present the simpler case in which only non-green production is taxed.

²We study increasing returns ($\psi > 0$) in the dynamic model; results are reported in Section 5.4.

Firms benefit from the production subsidy τ_g but will open vacancies only if they expect sufficient worker entry; workers face entry cost $(1 - s_g)\kappa_g$ and will invest only if they expect enough vacancies. Entry decisions on both sides are therefore mutually reinforcing. As we show below, this strategic complementarity generates multiple equilibria.

3.1.2 Workers

Unemployed workers receive z . If they enter the green market, they pay $(1 - s_g)\kappa_g$. Expected payoffs in each sector are:

$$\begin{aligned} \text{Green:} \quad & -(1 - s_g)\kappa_g + \alpha_{wg}w_g + (1 - \alpha_{wg})z, \\ \text{Non-green:} \quad & \alpha_{wn}w_n + (1 - \alpha_{wn})z. \end{aligned}$$

In equilibrium, workers are indifferent across markets,

$$-(1 - s_g)\kappa_g + \frac{v_g}{\pi + v_g}w_g + \left(1 - \frac{v_g}{\pi + v_g}\right)z = \frac{v_n}{(1 - \pi) + v_n}w_n + \left(1 - \frac{v_n}{(1 - \pi) + v_n}\right)z. \quad (3)$$

3.1.3 Firms

Free entry implies firms post vacancies until expected profits equal the recruiting cost c :

$$\begin{aligned} \text{Green:} \quad & c = \alpha_{fg}(p + \tau_g - w_g), \\ \text{Non-green:} \quad & c = \alpha_{fn}(p - \tau_n - w_n), \end{aligned}$$

where p is worker productivity (common across sectors). Subsidies and taxes shift the surplus split between workers and firms.

3.2 Wage Determination (Nash Bargaining)

With worker bargaining power η , the static Nash solution yields

$$w_g = \eta(p + \tau_g) + (1 - \eta)z, \quad w_n = \eta(p - \tau_n) + (1 - \eta)z.$$

Derivations are in Appendix A.

3.2.1 Equilibrium

An equilibrium consists of wages (w_g, w_n) , vacancies (v_g, v_n) , unemployment and employment masses (u_g, u_n, e_g, e_n) , the worker entry share π , and policy instruments (τ_g, τ_n, s_g) such that: (i) workers' indifference holds; (ii) firm free-entry conditions hold; (iii) matching probabilities satisfy (2); and (iv) market sizes add up to the unit labor force.

Proposition 1. *The static economy admits multiple equilibria with following properties³:*

- $\exists$ a corner equilibrium where $\pi = 0$ and there are no jobs/production in the economy is green.
- $\exists$ a corner equilibrium where $\pi = 1$ and all jobs/production in the economy is green.
- For CRS case with $\psi = 0$, $\lim_{\pi \rightarrow 0+} G(\pi) > 0 > G(0)$ and $\lim_{\pi \rightarrow 1-} G(\pi) < 0 < G(1)$, i.e. the corner equilibria are not robust to small trembles, but $\exists$ at least one robust interior equilibrium.

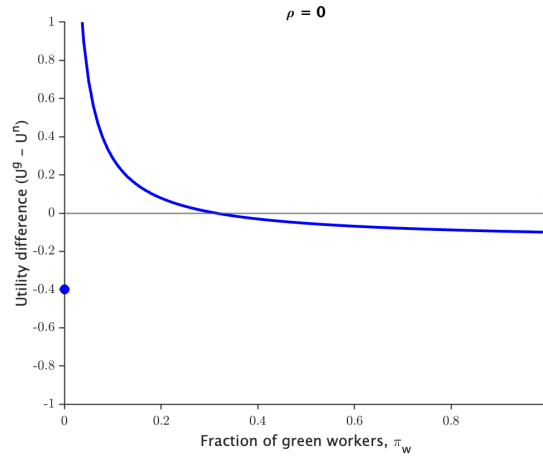


Figure 1. The reply function (G–N) for CRS ($\rho = 0$), showing multiplicity of equilibria in the static benchmark.

3.3 Comparative Statics

We conduct comparative statics in the static model to illustrate the main policy channels. Figure 2a shows that reducing workers' entry cost (higher s_g or lower κ_g) raises equilibrium green entry; Figure 2b shows that a higher firm subsidy τ_g likewise increases green entry.

³In a different context centered on asset liquidity, (Geromichalos, Herrenbrueck, and Lee, 2023) investigates how agents select among various asset markets when faced with random liquidity demands, deriving a solution that mirrors the characterization presented here.

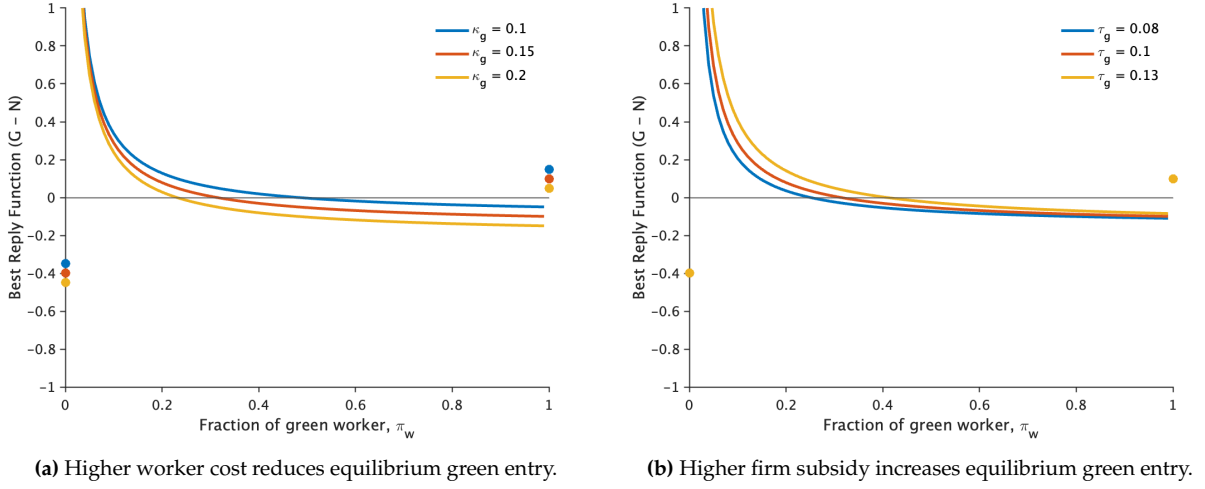


Figure 2. Comparative statics: entry costs and subsidies on equilibrium green entry.

Overall, the static benchmark highlights the need to incentivize *both* sides of the market. It also provides a clear framework for the efficiency analysis below, where we show how environmental externalities interact with the standard search externality and how policy instruments can decentralize the planner's allocation.

3.4 Social Efficiency and Policy Implications⁴

The decentralized equilibrium may be inefficient for two reasons: (i) standard search-and-matching externalities, and (ii) uninternalized environmental externalities from green job creation. We characterize efficiency using a social planner who internalizes both.

3.4.1 Planner's Problem Without Environmental Externalities

If green jobs yield no external benefits ($\phi = 0$), the planner maximizes total surplus:

$$\max_{\{\pi, v_g, v_n\}} L(\pi, v_g, v_n) = \underbrace{\pi \frac{v_g}{\pi + v_g} (p - z)}_{\text{Green matches}} + \underbrace{(1 - \pi) \frac{v_n}{(1 - \pi) + v_n} (p - z)}_{\text{Non-green matches}} - c(v_g + v_n) - \pi \kappa_g. \quad (4)$$

Proposition 2. *Without externalities ($\phi = 0$), the planner's solution is a corner: if $\kappa_g > 0$, then $\pi^* = 0$ and all workers enter the non-green sector. No interior allocation is efficient.*

Proof in Appendix B.1.

This result arises because the private cost of green entry κ_g dominates any productivity gains, making the non-green allocation efficient.

⁴All proofs and derivations are provided in Appendix B.

3.4.2 Planner's Problem With Environmental Externalities

When each green match generates an additional benefit $\phi > 0$ accruing to society, the planner maximizes:

$$\begin{aligned} \max_{\{\pi, v_g, v_n\}} L(\pi, v_g, v_n) = & \underbrace{\pi \frac{v_g}{\pi + v_g} [(p - z) + \phi]}_{\text{Green matches with externality}} + \underbrace{(1 - \pi) \frac{v_n}{(1 - \pi) + v_n} (p - z)}_{\text{Non-green matches}} \\ & - c(v_g + v_n) - \pi \kappa_g. \end{aligned} \quad (5)$$

In the decentralized equilibrium, this externality is not internalized by workers or firms, leading to underinvestment in green employment.

Proposition 3. *With $\phi > 0$, efficiency requires three alignment conditions:*

1. *Hosios condition: $\eta^{g*} = \frac{v_g^*}{\pi^* + v_g^*} = \eta^{n*} = \frac{v_n^*}{(1 - \pi^*) + v_n^*} = \eta$, i.e. bargaining power equals the matching elasticity in both sectors.*
2. *Firm-side alignment: $\phi = \frac{\tau_g + \tau_n}{1 - \eta}$.*
3. *Worker-side alignment: $\phi = \frac{\tau_g + \tau}{\eta(1 - s_g)}$.*

Proof in Appendix B.2.

In addition to the standard Hosios condition, efficiency requires that policy instruments (τ, τ_g, s_g) jointly align private and social returns for both firms and workers. Because these instruments are interdependent, not all combinations decentralize the planner's allocation. Comparative statics in the one-shot model illustrate these policy channels. To our knowledge, this is the first paper to characterize such efficiency conditions in a search-and-matching model with environmental externalities.

We now turn to the dynamic version of the model, which incorporates unemployment dynamics and policy financing over time.

3.5 Dynamic Model: Extended DMP Framework

Building on the static framework, the dynamic model introduces time and sectoral dynamics to analyze green labor market transitions more comprehensively. This extension permits a unified treatment of unemployment dynamics, funding requirements, and aggregate welfare—central issues for policy design in the green transition.

3.5.1 Physical Environment

Time is discrete, $t = 0, 1, 2, \dots$. Agents discount at rate $\beta \in (0, 1)$. The labor force is normalized to 1. The economy features two segmented labor markets, green (g) and non-green (n). Existing matches are destroyed at rate $\lambda \in (0, 1)$. The government provides a production subsidy τ_g to green firms financed by a tax τ_n on non-green firms. Workers entering the green sector incur a recurring entry cost κ_g , partially offset by a subsidy $s_g \in [0, 1]$; entry into the non-green sector is costless.

3.5.2 Matching Function

Matching in each sector $j \in \{g, n\}$ follows

$$f(u_j, v_j) = \delta_j \left(\frac{u_j v_j}{u_j + v_j} \right)^{1-\psi} (u_j v_j)^\psi, \quad \psi \in [0, 1],$$

where u_j and v_j denote unemployed workers and vacancies, and δ_j is sector-specific matching efficiency. In the constant-returns case ($\psi = 0$), the implied meeting probabilities are

$$\alpha_{wj} = \frac{v_j}{u_j + v_j}, \quad \alpha_{fj} = \frac{u_j}{u_j + v_j}, \quad j \in \{g, n\}. \quad (6)$$

3.5.3 Discussion of Modeling Choices and Empirical Relevance

Before we move on to the full quantitative analysis, we want to discuss some modeling choices.

Definition of green jobs: Green jobs are defined following [Curtis and Marinescu \(2022\)](#) as employment in renewable energy sectors (solar, wind, EVs), aligning with policy focus under the IRA and the fact that energy-related activities account for roughly 70% of U.S. anthropogenic emissions ([World Nuclear Association, 2024](#)). This operational definition complements broader notions (e.g., [International Labor Organization \(2024\)](#)) and task-based classifications (e.g., O*NET; [Vona, Marin, Consoli, and Popp, 2018](#); [Consoli, Marin, Marzucchi, and Vona, 2018](#); [Vona, Marin, and Consoli, 2019](#)) while capturing emerging occupations (e.g., solar installers).

Worker entry costs: κ_g represent the structural barriers faced by workers transitioning to green jobs. These costs are modeled broadly to reflect various frictions:

- *Training Costs:* Green jobs often demand new technical and managerial skills, requiring reskilling even in related occupations ([Vona et al., 2018](#); [Consoli et al., 2018](#)).

- *Relocation Costs*: Green jobs are geographically dispersed, necessitating relocation from fossil fuel hubs to regions rich in renewable resources (Brookings Institution, 2022; Johnson and Schulhofer-Wohl, 2019; Lim et al., 2023).
- *Unionization and Benefits*: Fossil fuel jobs typically offer stronger union representation and better benefits compared to green jobs, deterring workers from transitioning (Emden and Murphy, 2019; Pollin, Garrett-Peltier, et al., 2020).
- *Uncertainty and Behavioral Barriers*: Perceived instability in green jobs and behavioral factors such as risk aversion or sunk costs further impede transitions (Villas-Boas, 2021; Dixit and Rob, 1994).

Although the model remains agnostic about the exact composition of κ_g , its inclusion reflects the real-world barriers to green labor market flexibility and the economics of the model goes through any of those assumptions.

Firm subsidies, τ_g : The Inflation Reduction Act (IRA) provides substantial production and investment tax credits (τ_g) to incentivize renewable energy deployment, with credits ranging from \$5/MWh to \$32/MWh based on criteria such as labor conditions and domestic content (Bistline et al., 2023). These subsidies have accelerated green technology adoption but have largely overlooked workforce development (Walsh, 2023). By modeling τ_g , the framework captures the financial incentives driving renewable energy expansion while highlighting the gap in policies targeting the labor market.

Why policy-driven solutions? Unlike gradual, largely market-led transitions (e.g., automation or trade), the clean-energy transition is policy-anchored and time-bound. Meeting decarbonization goals requires rapid reallocation that can outpace organic adjustment and historical restructurings show the risks of such unmanaged shifts (Oei et al., 2020). Policy is thus essential to align incentives, overcome coordination frictions, and limit transitional unemployment.

3.5.4 Analysis of the Model

We derive equilibrium conditions linking policy instruments to Beveridge curves, value functions, and wages. This structure permits quantitative evaluation of green employment targets and the associated unemployment and fiscal paths.

Beveridge curves: Let u denote total unemployed workers who choose to enter either the green (u_g) or non-green (u_n) sector, where they may find a job and transition to employment (e_g and e_n , respectively). Similarly, employed workers in both sectors face the possibility of job destruction at rate λ , which returns them to unemployment. The worker-flow diagram in Figure 3 implies steady-state relationships:

$$\alpha_{wg}u_g = \lambda e_g, \quad \alpha_{wn}u_n = \lambda e_n, \quad u_g + u_n + e_g + e_n = 1. \quad (7)$$

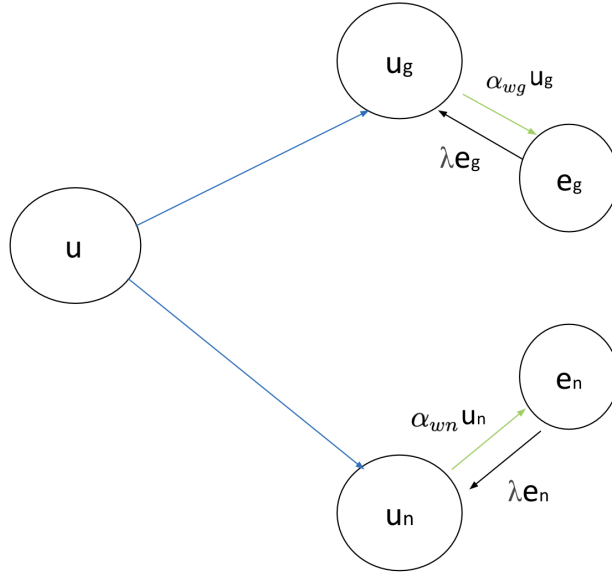


Figure 3. Worker flows between unemployment and employment in both sectors.

Firms' value functions and free entry: Firms choose between entering the green or non-green sectors based on expected profits, which is analogous to free entry in both sectors. Let V represent the value of a vacant firm, and $V_g(J_g)$ and $V_n(J_n)$ be the values of vacant (filled) jobs in the green and non-green sectors, respectively. Each value function is then:

$$V = \max\{V_g, V_n\}, \quad (8)$$

$$V_g = -c + \beta[\alpha_{fg}J_g + (1 - \alpha_{fg})V], \quad (9)$$

$$V_n = -c + \beta[\alpha_{fn}J_n + (1 - \alpha_{fn})V], \quad (10)$$

$$J_g = (p + \tau_g) - w_g + \beta[\lambda V + (1 - \lambda)J_g]^5, \quad (11)$$

$$J_n = p - \tau_n - w_n + \beta[(1 - \lambda)J_n + \lambda V]. \quad (12)$$

Free entry in both sectors implies $V_g = V_n = V = 0$, yielding

$$c = \beta\alpha_{fg}J_g, \quad c = \beta\alpha_{fn}J_n. \quad (13)$$

Imposing free entry to the value functions for filled firms also gives:

$$J_g = (p + \tau_g) - w_g + \beta(1 - \lambda)J_g, \quad J_n = p - w_n - \tau_n + \beta(1 - \lambda)J_n. \quad (14)$$

Workers' value functions and optimal sector choice: Let U represent the value of an unemployed worker, with $U_g(W_g)$ and $U_n(W_n)$ denoting the values of unemployed (employed) workers in the green and non-green sectors, respectively. Then the value functions are:

$$U = \max\{U_g, U_n\}, \quad (15)$$

$$U_g = z - (1 - s_g)\kappa_g + \beta[\alpha_{wg}W_g + (1 - \alpha_{wg})U_g], \quad (16)$$

$$U_n = z + \beta[\alpha_{wn}W_n + (1 - \alpha_{wn})U_n], \quad (17)$$

$$W_g = w_g + \beta[(1 - \lambda)W_g + \lambda U], \quad (18)$$

$$W_n = w_n + \beta[(1 - \lambda)W_n + \lambda U]. \quad (19)$$

Solving yields

$$U_g = \frac{[1 - \beta(1 - \lambda)](z - (1 - s_g)\kappa_g) + \beta\alpha_{wg}w_g}{(1 - \beta)(1 - \beta(1 - \alpha_{wg} - \lambda))}, \quad U_n = \frac{[1 - \beta(1 - \lambda)]z + \beta\alpha_{wn}w_n}{(1 - \beta)(1 - \beta(1 - \alpha_{wn} - \lambda))}.$$

In equilibrium, workers are indifferent across sectors:

$$U_g = U_n, \\ \implies \frac{[1 - \beta(1 - \lambda)](z - (1 - s_g)\kappa_g) + \beta\alpha_{wg}w_g}{(1 - \beta)(1 - \beta(1 - \alpha_{wg} - \lambda))} = \frac{[1 - \beta(1 - \lambda)]z + \beta\alpha_{wn}w_n}{(1 - \beta)(1 - \beta(1 - \alpha_{wn} - \lambda))}. \quad (20)$$

which pins down the endogenous composition of search across g and n .

⁵Since we normalize productivity p to 1 in the quantitative analysis, the green subsidy τ_g can be interpreted either as an additive transfer to firm revenue or equivalently as a per-unit output subsidy.

3.5.5 Bargaining Problems

Non-green sector: Let us start by describing the terms of trade in a meeting between a firm and an unemployed in the non-green sector. Solving the standard Nash bargaining problem leads to the following condition that must be satisfied:

$$(1 - \eta)(W_n - U_n) = \eta J_n, \quad (21)$$

This condition indicates that each party receives a share of the total surplus from the match, proportional to their bargaining power. (Recall that η represents the worker's bargaining power.) Substituting the value functions J_n , U_n , and W_n from above equations (12), (17), and (19) respectively, we can express the wage in the non-green sector as follows:

$$w_n = \frac{(1 - \eta)z[1 - \beta(1 - \lambda)] + \eta(p - \tau_n)[1 - \beta(1 - \lambda - \alpha_{wn})]}{1 - \beta(1 - \lambda - \eta\alpha_{wn})}. \quad (22)$$

(See Appendix C.1.) The wage depends on tightness through α_{wn} and falls with the tax τ_n .

Green sector: Next, consider the bargaining problem between a firm and a worker in the green sector, where workers incur real costs to enter the sector, while firms receive government subsidies when they match with workers and produce. Again, we have:

$$(1 - \eta)(W_g - U_g) = \eta J_g,$$

which yields

$$w_g = \frac{(1 - \eta)(z - (1 - s_g)\kappa_g)[1 - \beta(1 - \lambda)] + \eta(1 + \tau_g)p[1 - \beta(1 - \lambda - \alpha_{wg})]}{1 - \beta(1 - \lambda - \eta\alpha_{wg})}. \quad (23)$$

(See Appendix C.2.) The wage rises with firm- and worker-side support (τ_g and s_g) and declines with higher entry costs κ_g .

3.5.6 Government Budget Constraint

Tax revenues from the non-green sector finance subsidies to green firms and workers:

$$\underbrace{\tau_n \cdot \alpha_{fn} v_n}_{\text{Tax revenue from non-green firms}} = \underbrace{\tau_g \cdot \alpha_{fg} v_g}_{\text{Subsidies to green firms}} + \underbrace{s_g \cdot u_g}_{\text{Subsidies to green workers}}.$$

$$\tau_n = \frac{\tau_g \alpha_{fg} v_g + s_g u_g}{\alpha_{fn} v_n}, \quad (24)$$

where α_{fj} are firm matching rates and u_g, v_g, v_n denote the relevant stocks.

3.5.7 Definition of Steady-State Equilibrium

A steady state equilibrium in our model consists of wages (w_g, w_n) for workers in the green and non-green sectors, a green production subsidy τ_g , a flat tax τ_n paid by non-green firms, measures of vacancies in both sectors (v_g, v_n) , and measures of employed and unemployed workers in the various states (u_g, u_n, e_g, e_n) . The subsidy and tax satisfy the government budget constraint (24). The remaining equilibrium variables satisfy two free entry conditions (13), two wage curves (22, 23), three Beveridge curves (7), and optimal entry condition (20), after one replaces the various α 's with the respective matching probabilities given in (6).

4 Calibration

We calibrate the benchmark model to the U.S. economy in 2022, focusing on labor market dynamics in the green transition. One model period corresponds to one calendar month. Parameters with direct empirical counterparts are set exogenously. The discount factor is $\beta = 0.9959$, consistent with a 5% annual interest rate. Worker productivity p is normalized to 1, and the matching function is assumed to exhibit constant returns to scale ($\psi = 0$). Following Shimer (2005), workers' bargaining power is set to $\eta = 0.72$, and the non-employment benefit to $z = 0.4p$ (40% of average productivity).

Several calibration targets discipline the model. First, we target a 2% wage premium for green workers relative to non-green workers, based on IMF staff estimates (Fund, 2022).⁶ Second, the green employment share is set at 2% of the total U.S. workforce, consistent with 2022 estimates from the Energy Information Administration ((EIA), 2024) and calculation of employment share in the renewable sector⁷. Third, the hiring likelihood ratio for workers with green skills relative to others is set to 29%, as reported by LinkedIn's *Green Skills Report* (LinkedIn, 2023). Fourth, overall labor market tightness, measured as the ratio of vacancies to unemployed workers, is set to 1.868, consistent with FRED data

⁶Other studies find larger premiums, e.g., about 4% in VoxEU (2023) and up to 20% in Curtis and Marinescu (2022), but we adopt the conservative estimate.

⁷In 2022, there were 3.3 million renewable energy jobs ((E2), 2023) and 212.4 million total jobs ((REA), 2024), resulting in $e_g/e \approx 2\%$.

(FRED Blog, 2024). Finally, the unemployment rate is targeted at 3.5%, matching BLS data for 2023 (of Labor Statistics (BLS), 2023).

We also incorporate the scale of green subsidies under the Inflation Reduction Act (IRA). Production and investment tax credits range from \$5/MWh to \$32/MWh depending on eligibility (Bushnell and Smith, 2024). With 0.91 billion MWh of renewable generation in 2022 (U.S. Department of Energy, 2023) and GDP of \$25.44 trillion (Bank, 2022), these subsidies correspond to 0.018–0.114% of GDP. We set the benchmark subsidy rate at 0.1% of GDP.

The model matches the targeted moments closely, as shown in Table 1.

Moment	Model	Target
Wage premium (green/non-green)	1.018	1.020
Employment share (green)	0.0195	0.0195
Hiring likelihood ratio	1.29	1.29
Labor market tightness	1.868	1.868
Unemployment rate	3.5%	3.5%

Table 1: Model performance in matching calibration targets.

The internally calibrated parameters needed to match these moments are reported in Table 2.

Parameter	Description	Value
c	Vacancy creation cost	0.1640
λ	Separation rate	0.0188
δ_g	Matching efficiency (green)	0.8826
δ_n	Matching efficiency (non-green)	0.7961
κ_g	Worker entry cost (green)	0.7246

Table 2: Internally calibrated parameters.

This calibration anchors the model to key features of the U.S. labor market in 2022 and provides a benchmark for policy experiments. With the model successfully reproducing the wage premium, employment shares, hiring patterns, tightness, and unemployment rate, we are equipped to address the central policy question: *Which mix of worker-side and firm-side interventions can raise green employment from 2% to 14% of the labor force by 2030, as projected by WorkingNation (2024), while minimizing unemployment costs and fiscal burdens?*

5 Quantitative Analysis

The U.S. aims to raise the share of green employment from 2% to 14% of total jobs by 2030, implying nearly 24 million green jobs (WorkingNation, 2024). Using the calibrated model, we evaluate the policy levers needed to achieve this target, focusing on two margins: (i) lowering worker entry costs into the green sector (κ_g , via subsidies s_g) and (ii) raising production subsidies for green firms (τ_g).

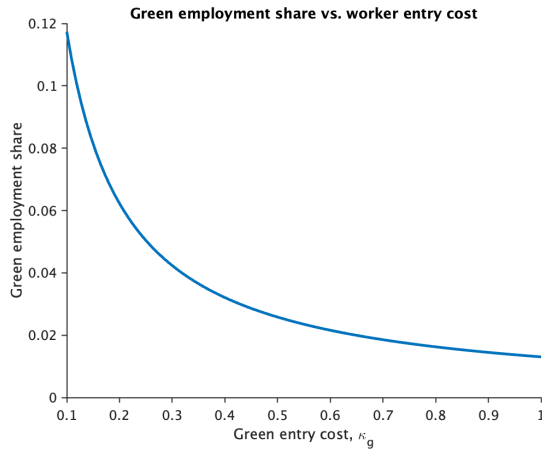


Figure 4. Lower entry costs raise green employment share.

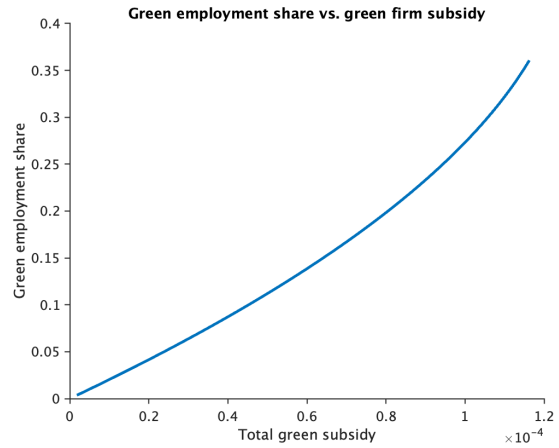


Figure 5. Higher firm subsidies raise green employment share.

5.1 Reaching the Employment Target

Figure 4–5 show that both worker- and firm-side policies independently increase the equilibrium green employment share. Table 3 compares strategies that combine these margins in different ways. All strategies can, in principle, achieve the 14% target, but they differ in cost and efficiency.

Table 3: Alternative Policy Paths to 14% Green Employment

Equilibrium	Per worker cost subsidy	Per firm green subsidy	Total green firm subsidy	Green emp. share
Dual: worker + firm support	0.8895	0.2565	6.45e-04	14%
Fix per firm subsidy, reduce worker cost	0.9056	0.0272	6.80e-05	14%
Fix total firm subsidy, reduce worker cost	0.9002	0.0036	8.91e-06	14%
Fix worker cost, raise firm subsidy	0	2.5273	0.0063	14%
Baseline (2022)	0	0.0272	8.91e-06	2%

The baseline economy has only 2% of jobs in the green sector. The four strategies in Table 3 show that both worker- and firm-side interventions are individually effective, but

the combined approach is quantitatively different: it achieves the employment goal with substantially lower subsidies than a one-sided policy.

5.2 Welfare, Funding, and Unemployment

Table 4 compares alternative policy strategies along three margins: aggregate welfare, funding requirements, and unemployment. The dual strategy—lowering worker entry costs while raising firm subsidies—dominates, delivering the lowest unemployment, smallest funding burden, and highest welfare among feasible policies.

Table 4: Policy Tradeoffs: Welfare, Funding, and Unemployment

Equilibrium	Welfare	Funding Req.	Unemployment
Dual: worker + firm support	0.9617	0.00480	0.0450
Fix per firm subsidy, lower worker cost	0.9598	0.00517	0.0476
Fix total firm subsidy, lower worker cost	0.9597	0.00519	0.0478
Fix worker cost, raise firm subsidy	0.9554	0.00630	0.0533
Baseline (2022)	0.9675	8.91e-06	0.0350

Relative to worker-side-only subsidies, the dual approach raises welfare by 0.2%, cuts required funding by about 8%, and reduces unemployment by 6%. Compared to a firm-only subsidy, the advantages are larger: welfare is 0.7% higher, funding needs are 31% lower, and unemployment falls by 18%. In summary, reducing worker entry costs and increasing green subsidies simultaneously is the most efficient policy combination, maximizing welfare, minimizing funding burdens, and substantially lowering unemployment.

The striking result is that all policy combinations still yield lower welfare than the baseline (0.9675). This arises because, in the baseline DMP model, green jobs provide no external benefits. Purely policy-driven reallocation thus lowers average productivity, even as employment goals are met. This motivates the next extension.

5.3 Welfare with Green Externalities

We incorporate positive externalities from green employment into the welfare function. The standard DMP welfare function is given by:

$$W_0 = p(e_g + e_n) + (z - \kappa_g)u_g + zu_n - c(v_n + v_g).$$

Adding a per-job externality ϕ , capturing the environmental benefit generated by each unit of employment in the green sector, yields

$$W_\phi = p(e_g + e_n) + (z - \kappa_g)u_g + zu_n - c(v_n + v_g) + \phi e_g.$$

5.3.1 Threshold Analysis

The welfare gain or loss from policy interventions depends critically on the value of ϕ . Figure 6 shows how welfare depends on ϕ . Under the dual policy, welfare equals the baseline ($W_\phi = W_0$) when $\phi = 0.0432$. Thus, only modest externalities (bigger than $\phi = 0.0432$) are required for green reallocation to become welfare-improving.

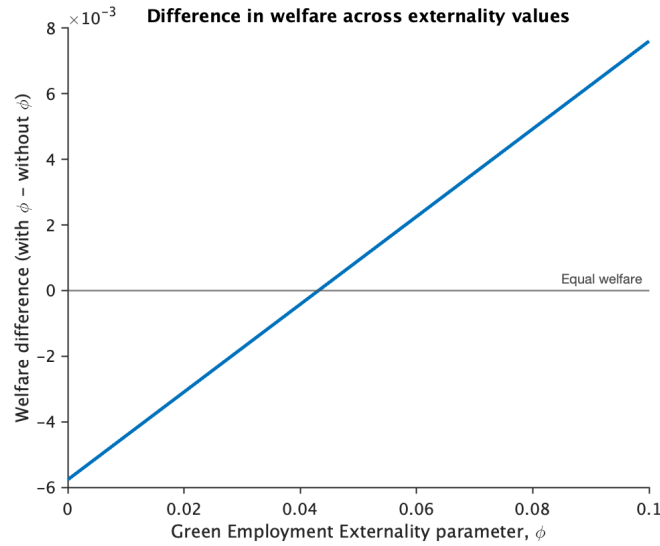


Figure 6. Welfare gains as a function of externality parameter ϕ .

5.3.2 Productivity Equivalence of the Externality Threshold

To interpret the welfare threshold $\phi = 0.0432$, we translate it into productivity units. Specifically, we ask: *by how much would worker productivity p need to rise to generate the same welfare gain as this externality?* The calculation shows that $\phi = 0.0432$ is equivalent to a 0.6% increase in aggregate productivity. Put differently, expanding the green sector becomes welfare-improving once its environmental benefits are at least as valuable as a 0.6% productivity gain. This provides a simple benchmark for quantifying externalities and embedding them into DMP-style labor market models.

5.4 Increasing Returns to Scale in Matching

Finally, we consider the case where the matching function exhibits mild increasing returns to scale (IRS), $\psi > 0$. Table 5 shows that even small departures from CRS amplify the feedback between firm vacancies and worker entry, lowering unemployment, raising welfare, and reducing subsidy needs.

Table 5: Effects of IRS on Unemployment, Welfare, and Subsidy Needs

$\psi_n = \psi_g$	Unemployment ↓	Welfare ↑	Subsidy ↓
0.000	0.0450	0.9617	0.00480
0.001	0.0429	0.9628	0.00434
0.010	0.0408	0.9641	0.00368
0.020	0.0409	0.9644	0.00363

IRS magnifies the payoff of a larger (worker, firm) pool in one sector. This dynamic further supports the notion that coordinated policies targeting both sides of the market can quickly shift the system away from low-green equilibria.

Summary: Quantitative analysis yields three main insights. First, worker- and firm-side policies can individually achieve the 14% employment target, but coordinated interventions are far more efficient. Second, modest positive externalities are sufficient for these policies to deliver net welfare gains. Third, mild increasing returns to scale amplify these dynamics, further lowering unemployment and subsidy needs.

6 Economic Insights

The analysis shows that a *dual policy strategy* most effectively resolves the coordination failure that slows green labor reallocation. Worker-side subsidies reduce barriers to entry and training, while firm-side subsidies lower hiring costs and expand labor demand. When implemented together, these policies reinforce each other, improving matching efficiency, lowering fiscal costs, and maximizing welfare. By contrast, one-sided interventions fail to address the interdependence between firms and workers, requiring much larger fiscal outlays to achieve the same employment target and generating greater unemployment along the way.

Allowing for even *mild increasing returns to scale* in matching magnifies this dynamic. Once a critical mass of workers and firms enter the green sector, job matching accelerates endogenously, reducing fiscal costs and further strengthening incentives for both

sides to go green. Yet, absent any environmental externality, reallocating labor into the green sector still lowers measured welfare relative to the baseline, since the model only counts productivity and fiscal costs. Introducing a modest positive externality from green employment reverses this result. Thus, properly valuing these externalities is therefore essential to assess the true net benefits of green labor policies.

7 Conclusion

This paper highlights the central role of policy design in shaping the efficiency of large-scale labor reallocations. Achieving net-zero targets is one example, but the lessons apply more broadly to structural shifts such as digitalization, demographic change, or trade realignment. In all cases, the challenge is not simply to move workers across sectors, but to do so in a way that avoids unnecessary unemployment, minimizes fiscal costs, and mitigates welfare loss.

We illustrate this with the U.S. green transition using an extended DMP model framework. One-sided interventions—targeting only firms or only workers—leave coordination frictions unresolved: firms delay vacancies without a trained labor pool, while workers hesitate to retrain without credible opportunities. The result is higher unemployment, larger fiscal burdens, and lower welfare. By contrast, a coordinated dual strategy breaks this chicken-and-egg problem, raising green employment from 2% to 14% by 2030 while lowering unemployment and funding needs. Mild increasing returns to scale reinforce these gains, and incorporating even modest environmental externalities (equivalent to a 0.6% productivity boost) makes the transition welfare-improving relative to the baseline.

The policy implication is clear: to manage structural transitions in a fiscally sustainable and socially efficient way, governments must treat workers and firms as co-dependent levers, not isolated targets. More broadly, the framework offers a tool for analyzing other major transitions across advanced and emerging economies, where factors of production must shift quickly but frictions risk stalling adjustment.

References

- Autor, D. H., D. Dorn, and G. H. Hanson (2013). The china syndrome: Local labor market effects of import competition in the united states. *American economic review* 103(6), 2121–2168.
- Bank, W. (2022). Gdp (current us\$) - united states. Accessed: 2024-10-18.
- BCG, B. C. G. (2023). To reach sustainability goals, organizations must address critical sustainability skills gap. BCG Henderson Institute.
- Bilal, A. and J. H. Stock (2025). Macroeconomics and climate change.
- Bistline, J. E. et al. (2023). Economic impacts of the inflation reduction act on u.s. energy and climate policy. *Energy Economics*.
- Bluedorn, J., N.-J. Hansen, D. Noureldin, I. Shibata, and M. M. Tavares (2023). Transitioning to a greener labor market: Cross-country evidence from microdata. *Energy Economics* 126, 106836.
- Brookings Institution (2022). How renewable energy jobs can uplift fossil fuel communities and remake climate politics.
- Brown, C. (2015). Green transitions: Initial capital costs and financial challenges for firms. *Journal of Environmental Economics and Management* 71, 143–158.
- Bushnell, J. B. and A. Smith (2024). Modeling uncertainty in climate policy: An application to the us ira. Technical report, National Bureau of Economic Research.
- Böhringer, C., N. Rivers, and T. F. Rutherford (2013). Climate policy and inequality: An assessment of the social impact of climate policy on income distribution. *Journal of Environmental Economics and Management* 66(2), 211–228.
- Card, D., J. Kluve, and A. Weber (2018). What works? a meta analysis of recent active labor market program evaluations. *Journal of Economic Literature* 56(3), 905–976.
- Consoli, D., G. Marin, A. Marzucchi, and F. Vona (2018). Green skills. *Journal of Environmental Innovation and Societal Transitions* 10, 86–101.
- Conte, B., K. Desmet, and E. Rossi-Hansberg (2022). On the geographic implications of carbon taxes. NBER Working Paper 30678, National Bureau of Economic Research.
- Cooper, R. and A. John (1988). Coordinating coordination failures in keynesian models. *The Quarterly Journal of Economics* 103(3), 441–463.
- Curtis, E. M. and I. Marinescu (2022). Green energy jobs in the us: What are they, and where are they? Technical report, National Bureau of Economic Research.
- Diamond, P. A. (1982). Aggregate demand management in search equilibrium. *The Journal of Political Economy* 90(5), 881–894.
- Dixit, A. and R. Rob (1994). Switching costs and sectoral shifts: Evidence from microdata.

- Quarterly Journal of Economics* 109(4), 1387–1412.
- (E2), E. E. (2023). Clean jobs america 2023. Accessed: 2024-10-18.
- (EIA), U. E. I. A. (2024). Annual energy outlook. Accessed: 2024-10-18.
- Emden, J. and L. Murphy (2019). Power struggle: Fossil fuel jobs, green jobs, and the energy transition. IPPR Environmental Research Report.
- FRED Blog, F. R. B. o. S. L. (2024). The job openings to unemployment ratio: Labor markets are in better balance. Accessed: 2024-10-18.
- Freire-González, J. (2018). Environmental taxation and the double dividend hypothesis in cge models: A critical review. *Journal of Policy Modeling* 40(1), 194–223.
- Fund, I. M. (2022). How the green transition will impact u.s. jobs. Accessed: 2024-10-18.
- Geromichalos, A., L. Herrenbrueck, and S. Lee (2023). The strategic determination of the supply of liquid assets. *Review of Economic Dynamics* 49, 1–36.
- Gibson, M. and G. Heutel (2023). Environmental policy and business cycles with labor market frictions. *Journal of Environmental Economics and Management* 107, 102–118.
- Hafstead, M. A. C. and R. C. Williams III (2018). Environmental policy, full-employment models, and employment: The role of labor market frictions. *Journal of the Association of Environmental and Resource Economists* 5(3), 673–709.
- Hafstead, M. A. C., R. C. Williams III, and Y. Chen (2022). Labor market frictions, green jobs, and the energy transition. *Resource and Energy Economics* 70, 101–132.
- ILO, I. L. O. (2019). *Skills for a Greener Future: A Global View*. Geneva: International Labor Organization.
- International Labor Organization (2024). What is a green job?
- Johnson, J. and S. Schulhofer-Wohl (2019). Uneven job growth and geographic mobility. *Journal of Labor Economics* 37(1), 105–133.
- Kluve, J., S. Puerto, D. Robalino, J. M. Romero, F. Rother, J. Stöterau, F. Weidenkaff, and M. Witte (2019). Do youth employment programs improve labor market outcomes? a quantitative review. *Journal of Development Effectiveness* 11(5), 545–566.
- Lankhuizen, M., H. Rojas-Romagosa, and C. van Ewijk (2022). Matching jobs with skills during the energy transition: Labor market bottlenecks in the netherlands. *Energy Economics* 106, 104705.
- Lim, S. et al. (2023). Spatial mismatch, relocation costs, and employment in transitioning labor markets. *American Economic Journal: Applied Economics*.
- LinkedIn (2023). Global green skills report 2023. Accessed: 2024-10-18.
- LinkedIn (2024). Global climate talent stocktake. Accessed: 2024-10-14.
- Mortensen, D. T. and C. A. Pissarides (1994). Job creation and job destruction in the theory of unemployment. *The Review of Economic Studies* 61(3), 397–415.

- Oei, P.-Y. et al. (2020). Lessons learned from the coal phase-out in germany: Policies and transition pathways for a sustainable future. *Energy Policy* 146, 1111–1120.
- of Labor Statistics (BLS), U. B. (2023). Current population survey (cps). Accessed: 2024-10-18.
- Patuelli, R., P. Nijkamp, and E. Pels (2005). Environmental externalities in regional economic models: Concepts, issues and methods. *Environment and Planning A* 37(5), 845–863.
- Pissarides, C. A. (2000). *Equilibrium Unemployment Theory* (2nd ed.). Cambridge, MA: MIT Press.
- Pollin, R., H. Garrett-Peltier, et al. (2020). Green recovery: Assessing the employment impacts of green energy investments for the u.s. *Political Economy Research Institute Report*.
- (REA), R. E. A. P. (2024). Total employment comparative trends analysis. Accessed: 2024-10-18.
- Sancho, F. (2010). Green tax reforms and double dividend in decentralized economies. *Journal of Environmental Economics and Management* 59(1), 42–53.
- Shapiro, A. F. and G. E. Metcalf (2023). The macroeconomic effects of a carbon tax to meet the us paris agreement target: The role of firm creation and technology adoption. *Journal of Public Economics* 218, 104800.
- Shimer, R. (2005). The cyclical behavior of equilibrium unemployment and vacancies. *American economic review*, 25–49.
- U.S. Department of Energy (2023). Fotw #1291, may 22, 2023: Net generation of electricity from renewable sources exceeded coal for the first time in 2022. Accessed: 2024-10-18.
- Villas-Boas, S. B. (2021). Uncertainty in green job transitions: A labor market analysis.
- Vona, F., G. Marin, and D. Consoli (2019). Measures, drivers, and effects of green employment: Evidence from us local labor markets, 2006–2014. *Journal of Economic Geography* 19(5), 1021–1048.
- Vona, F., G. Marin, D. Consoli, and D. Popp (2018). Environmental regulation and the greening of jobs: An empirical analysis of the labor market transition. *Journal of the Association of Environmental and Resource Economists* 5(4), 865–891.
- VoxEU (2023). Labor markets and the green economy. *VoxEU*. Accessed: 2024-10-18.
- Walker, W. R. (2011). Environmental regulation and labor reallocation: Evidence from the clean air act. *American Economic Review* 101(3), 442–447.
- Walker, W. R. (2013). The transitional costs of sectoral reallocation: Evidence from the clean air act and the workforce. *The Quarterly Journal of Economics* 128(4), 1787–1835.
- Walsh, J. (2023). Bluegreen alliance statement on ira’s training and skills gaps.

WorkingNation (2024). Green jobs now: Opportunities for a clean energy future. Accessed: October 2, 2024.

World Nuclear Association (2024). Energy and emissions in the us.

A One-shot DMP Model

This section provides the detailed derivations for the static (one-period) benchmark version of the model, which is useful to illustrate the role of coordination frictions and the operation of policy instruments in segmented labor markets.

A.1 Bargaining

Green Sector: In the green sector, wages are determined via Nash bargaining between firms and workers. The problem is

$$\begin{aligned} & \max_{w_g} (w_g - z)^\eta (p + \tau_g - w_g)^{1-\eta}, \\ & \implies \frac{\eta}{w_g - z} = \frac{1 - \eta}{p + \tau_g - w_g}, \\ & \therefore w_g = \eta(p + \tau_g) + (1 - \eta)z. \end{aligned}$$

The bargained wage in the green sector is increasing in productivity p and the production subsidy τ_g , while the fallback option z provides the lower bound.

Non-Green Sector: In the non-green sector, the structure of the bargaining problem is analogous, except that firms pay a per-unit tax τ_n :

$$\begin{aligned} & \max_{w_n} (w_n - z)^\eta (p - \tau_n - w_n)^{1-\eta}, \\ & \implies \frac{\eta}{w_n - z} = \frac{1 - \eta}{p - \tau_n - w_n}, \\ & \therefore w_n = \eta(p - \tau_n) + (1 - \eta)z. \end{aligned}$$

Relative to the green sector, non-green wages are depressed by the tax burden τ_n .

A.2 Interior Equilibrium with Both Sectors Operating

When both sectors are active, the equilibrium variables are $\{\pi, v_g, v_n, w_g, w_n\}$, where π is the share of workers entering the green sector. The conditions are:

$$\begin{aligned} c &= \frac{\pi}{\pi + v_g}(p + \tau_g - w_g), \\ c &= \frac{1 - \pi}{1 - \pi + v_n}(p - \tau_n - w_n), \\ w_g &= \eta(p + \tau_g) + (1 - \eta)z, \\ w_n &= \eta(p - \tau_n) + (1 - \eta)z, \\ \tau_n \frac{(1 - \pi)}{(1 - \pi) + v_n} v_n &= \tau_g \frac{\pi}{\pi + v_g} v_g + s_g \pi \kappa_g, \\ \pi &= \begin{cases} 0 & \text{if } G < N, \\ \in (0, 1) & \text{if } G = N, \\ 1 & \text{if } G > N, \end{cases} \end{aligned}$$

where the expected utilities of workers are

$$G = -(1 - s_g)\kappa_g + \frac{v_g}{\pi + v_g}w_g + \left(1 - \frac{v_g}{\pi + v_g}\right)z, \quad N = \frac{v_n}{1 - \pi + v_n}w_n + \left(1 - \frac{v_n}{1 - \pi + v_n}\right)z.$$

A.3 Corner Equilibria

The economy can also settle in corner solutions.

A.3.1 Case $\pi = 0$: Non-Green Only

It must be the case that $v_g = 0$ iff $\pi = 0$. In this case, only non-green labor market operates with $v_n > 0$ and the matching probabilities simplify to

$$\alpha_{wg} = \alpha_{fg} = 0, \quad \alpha_{wn} = \frac{v_n}{1 + v_n}, \quad \alpha_{fn} = \frac{1}{1 + v_n}.$$

Equilibrium conditions are

$$\begin{aligned}
v_g &= 0, \\
c &= \frac{1}{1+v_n}(p - \tau_n - w_n), \\
w_g &= \eta(p + \tau_g) + (1 - \eta)z, \\
w_n &= \eta(p - \tau_n) + (1 - \eta)z, \\
\tau_n &= \tau_g = 0, \\
\pi &= 0 \text{ with } G < N,
\end{aligned}$$

where G and N are the expected utilities in the green and non-green sectors, respectively:

$$G = -\kappa_g + z, \quad N = \frac{v_n}{1+v_n}w_n + \left(1 - \frac{v_n}{1+v_n}\right)z, \quad G - N = -(1-s_g)\kappa_g - \frac{v_n}{1+v_n}(w_n - z) < 0.$$

A.3.2 Case $\pi = 1$: Green Only

It must be the case that $v_n = 0$ iff $\pi = 1$. In this case, only green labor market operates with $v_g > 0$ and the matching probabilities are:

$$\alpha_{wn} = \alpha_{fn} = 0, \quad \alpha_{wg} = \frac{v_g}{1+v_g}, \quad \alpha_{fg} = \frac{1}{1+v_g}.$$

Equilibrium conditions are:

$$\begin{aligned}
v_n &= 0, \\
c &= \frac{1}{1+v_g}(p + \tau_g - w_g), \\
w_g &= \eta(p + \tau_g) + (1 - \eta)z, \\
w_n &= \eta(p - \tau_n) + (1 - \eta)z, \\
\tau_n &= \tau_g = 0, \\
\pi &= 1 \text{ with } G > N,
\end{aligned}$$

with

$$G = -\kappa_g + \frac{v_g}{1+v_g}w_g + \left(1 - \frac{v_g}{1+v_g}\right)z, \quad N = z, \quad G - N = -(1-s_g)\kappa_g + \frac{v_g}{1+v_g}(w_g - z) > 0.$$

B Social Planner's Problem and Efficiency

B.1 Planner without Externality

We begin by considering a benchmark case without environmental externalities ($\phi = 0$), where the only difference between green and non-green sectors is that green-sector workers must pay an entry cost κ_g . The planner maximizes

$$L(\pi, v_g, v_n) = \pi \frac{v_g}{\pi + v_g} (p - z) + (1 - \pi) \frac{v_n}{(1 - \pi) + v_n} (p - z) - c(v_g + v_n) - \pi \kappa_g. \quad (\text{A.1})$$

The FOCs yield

$$\frac{\pi}{\pi + v_g} = \sqrt{\frac{c}{p - z}}, \quad \frac{1 - \pi}{1 - \pi + v_n} = \sqrt{\frac{c}{p - z}},$$

so match probabilities are equal across sectors. The worker allocation condition is

$$\left(\frac{v_g}{\pi + v_g} \right)^2 (p - z) - \left(\frac{v_n}{1 - \pi + v_n} \right)^2 (p - z) = \kappa_g.$$

This cannot hold with equal probabilities unless $\kappa_g = 0$. Thus the planner chooses a corner π .

B.2 Planner with Positive Externality ($\phi > 0$)

Now green matches generate an external benefit ϕ . With $\phi > 0$, the planner maximizes:

$$\max_{\{\pi, v_g, v_n\}} L(\pi, v_g, v_n) = \underbrace{\pi \frac{v_g}{\pi + v_g} [(y - z) + \phi]}_{\text{Green matches with externality } \phi > 0} + \underbrace{(1 - \pi) \frac{v_n}{(1 - \pi) + v_n} (y - z)}_{\text{non-green matches}} - c(v_g + v_n) - \pi \kappa_g,$$

FOCs imply efficient match probabilities

$$\frac{v_g}{\pi + v_g} = 1 - \sqrt{\frac{c}{(p - z) + \phi}}, \quad \frac{v_n}{1 - \pi + v_n} = 1 - \sqrt{\frac{c}{p - z}}.$$

The worker condition is

$$\left(\frac{v_g}{\pi + v_g} \right)^2 [(p - z) + \phi] - \left(\frac{v_n}{1 - \pi + v_n} \right)^2 (p - z) = \kappa_g.$$

This yields an interior solution as long as ϕ is large enough relative to κ_g .

B.3 Competitive Market with Policy

In equilibrium workers are indifferent if

$$-(1 - s_g)\kappa_g + \frac{v_g}{\pi + v_g}(w_g - z) = \frac{v_n}{1 - \pi + v_n}(w_n - z).$$

Wages are Nash outcomes:

$$w_g = \eta(p + \tau_g) + (1 - \eta)z, \quad w_n = \eta(p - \tau_n) + (1 - \eta)z.$$

Firms enter until

$$c = \frac{\pi}{\pi + v_g}(p + \tau_g - w_g), \quad c = \frac{1 - \pi}{1 - \pi + v_n}(p - \tau_n - w_n).$$

Implied green employment is

$$E_g = \pi \frac{v_g}{\pi + v_g} = \pi \left(1 - \frac{c}{(1 - \eta)(p + \tau_g - z)} \right).$$

Worker entry is pinned down by

$$\pi^* : \kappa_g = \frac{\eta(\tau_g + \tau_n)}{1 - s_g}.$$

B.4 Efficiency Conditions

To decentralize the planner's solution, the competitive equilibrium must replicate the planner's allocation $\{\pi^*, v_g^*, v_n^*\}$. We compare the worker and firm indifference conditions under both frameworks.

1. Firm Indifference Condition

Planner's condition:

$$c = \left[\frac{\pi}{\pi + v_g} \left(1 - \frac{v_g}{\pi + v_g} \right) - \frac{1 - \pi}{1 - \pi + v_n} \left(1 - \frac{v_n}{1 - \pi + v_n} \right) \right] (y - z) + \phi \cdot \frac{\pi}{\pi + v_g} \left(1 - \frac{v_g}{\pi + v_g} \right) = 0.$$

Market condition:

$$c = \left[\frac{\pi}{\pi + v_g} (1 - \theta) - \frac{1 - \pi}{1 - \pi + v_n} (1 - \theta) \right] (y - z) + \frac{\pi}{\pi + v_g} \tau_g + \frac{1 - \pi}{1 - \pi + v_n} \tau = 0.$$

2. Worker Indifference Condition

Planner's condition:

$$\kappa_g = \left[\left(\frac{v_g}{\pi + v_g} \right)^2 ((y - z)) - \left(\frac{v_n}{(1 - \pi) + v_n} \right)^2 (y - z) \right] + \phi \cdot \left(\frac{v_g}{\pi + v_g} \right)^2.$$

Market condition:

$$\kappa_g = \frac{\theta(\tau + \tau_g)}{1 - s_g}.$$

Conditions for Decentralization

To ensure efficiency (i.e., the planner's allocation is a competitive equilibrium), we require both the workers and firms indifference condition to coincide and the conditions are:

1. Hosios condition in both sector: $\eta^{g*} = \frac{v_g^*}{\pi^* + v_g^*} = \eta^{n*} = \frac{v_n^*}{1 - \pi^* + v_n^*} = \theta$.
2. Firm-side alignment: $\phi = \frac{\tau_g + \tau_n}{1 - \theta}$.
3. Workers-side alignment: $\phi = \frac{\tau + \tau_g}{\theta(1 - s_g)}$.

In addition to the standard Hosios condition, we now also have alignment conditions for both firms and workers. These conditions must be satisfied for a feasible set of policy instruments (τ, τ_g, s_g) to decentralize the efficient allocation. Importantly, note that the policy parameters are interdependent and not all combinations are feasible.

B.5 Government Budget

In steady state, total taxes collected from non-green-sector matches must cover the cost of green subsidies:

$$\tau_n \cdot \frac{(1 - \pi)}{(1 - \pi) + v_n} v_n = \tau_g \cdot \frac{\pi}{\pi + v_g} v_g + s_g \pi \kappa_g.$$

This constraint ties together the fiscal viability of the subsidy design with the labor market allocation.

C Bargaining in the Dynamic Model

This section derives the dynamic Nash bargaining solutions for wages in the non-green and green sectors. We begin from the value functions defined in the main text and proceed step-by-step to obtain closed-form expressions for the wage schedules. Throughout,

we maintain the notation introduced earlier: worker bargaining power is $\eta \in (0, 1)$, productivity is p , green firm subsidy is τ_g , non-green tax is τ_n , worker green-entry subsidy is s_g , worker green-entry cost is κ_g , discount factor is $\beta \in (0, 1)$, and the exogenous separation rate is $\lambda \in (0, 1)$. Matching probabilities are α_{wj} for workers and α_{fj} for firms in sector $j \in \{g, n\}$.

C.1 Non-Green Jobs

Proof. The non-green sector value functions (see equations (17), (19), and (12) in the main text) are:

$$U_n = z + \beta[\alpha_{wn}W_n + (1 - \alpha_{wn})U_n], \quad W_n = w_n + \beta[(1 - \lambda)W_n + \lambda U_n],$$

$$J_n = (p - \tau_n - w_n) + \beta[(1 - \lambda)J_n + \lambda V],$$

where V is the value of a vacancy and equals zero in equilibrium by free entry. Solving the first two linear equations yields expressions for U_n and W_n in terms of w_n and primitives.

Step 1: Solving for U_n . From the law of motion for U_n ,

$$U_n = z + \beta\alpha_{wn}W_n + \beta(1 - \alpha_{wn})U_n \implies U_n(1 - \beta(1 - \alpha_{wn})) = z + \beta\alpha_{wn}W_n,$$

so

$$U_n = \frac{z}{1 - \beta(1 - \alpha_{wn})} + \frac{\beta\alpha_{wn}}{1 - \beta(1 - \alpha_{wn})} W_n. \quad (\text{A.2})$$

Step 2: Solving for W_n . From the law of motion for W_n ,

$$W_n = w_n + \beta(1 - \lambda)W_n + \beta\lambda U_n \implies W_n(1 - \beta(1 - \lambda)) = w_n + \beta\lambda U_n.$$

Substitute (A.2):

$$W_n(1 - \beta(1 - \lambda)) = w_n + \beta\lambda \left[\frac{z}{1 - \beta(1 - \alpha_{wn})} + \frac{\beta\alpha_{wn}}{1 - \beta(1 - \alpha_{wn})} W_n \right].$$

Collecting terms in W_n ,

$$W_n \left(1 - \beta(1 - \lambda) - \frac{\beta^2\lambda\alpha_{wn}}{1 - \beta(1 - \alpha_{wn})} \right) = w_n + \frac{\beta\lambda}{1 - \beta(1 - \alpha_{wn})} z.$$

Multiply numerator and denominator to combine:

$$W_n \frac{(1 - \beta)(1 - \beta(1 - \alpha_{wn} - \lambda))}{1 - \beta(1 - \alpha_{wn})} = w_n + \frac{\beta\lambda}{1 - \beta(1 - \alpha_{wn})}z.$$

Therefore,

$$W_n = \frac{w_n(1 - \beta(1 - \alpha_{wn})) + \beta\lambda z}{(1 - \beta)(1 - \beta(1 - \alpha_{wn} - \lambda))}. \quad (\text{A.3})$$

Step 3: Substituting back into U_n . Using (A.2) and (A.3),

$$U_n = \frac{z}{1 - \beta(1 - \alpha_{wn})} + \frac{\beta\alpha_{wn}}{1 - \beta(1 - \alpha_{wn})} \cdot \frac{w_n(1 - \beta(1 - \alpha_{wn})) + \beta\lambda z}{(1 - \beta)(1 - \beta(1 - \alpha_{wn} - \lambda))}.$$

After simplification,

$$U_n = \frac{(1 - \beta(1 - \lambda))z + \beta\alpha_{wn}w_n}{(1 - \beta)(1 - \beta(1 - \alpha_{wn} - \lambda))}. \quad (\text{A.4})$$

Step 4: Surplus on the worker side. Subtract (A.4) from (A.3):

$$W_n - U_n = \frac{w_n - z}{1 - \beta(1 - \alpha_{wn} - \lambda)}.$$

This compact expression will be useful in the Nash condition.

Step 5: Firm surplus and Nash bargaining. With free entry, $V = 0$, so from the law of motion for J_n ,

$$J_n = (p - \tau_n - w_n) + \beta(1 - \lambda)J_n \implies J_n = \frac{p - \tau_n - w_n}{1 - \beta(1 - \lambda)}.$$

The Nash condition in the non-green sector is

$$(1 - \eta)(W_n - U_n) = \eta J_n.$$

Substitute the two surplus objects:

$$(1 - \eta) \frac{w_n - z}{1 - \beta(1 - \alpha_{wn} - \lambda)} = \eta \frac{p - \tau_n - w_n}{1 - \beta(1 - \lambda)}.$$

Solve for w_n :

$$w_n \left[\frac{1 - \eta}{1 - \beta(1 - \alpha_{wn} - \lambda)} + \frac{\eta}{1 - \beta(1 - \lambda)} \right] = \frac{(1 - \eta)z}{1 - \beta(1 - \alpha_{wn} - \lambda)} + \frac{\eta(p - \tau_n)}{1 - \beta(1 - \lambda)}.$$

After collecting terms over the common denominator $1 - \beta(1 - \lambda - \eta\alpha_{wn})$, we obtain

$$w_n = \frac{(1 - \eta)z[1 - \beta(1 - \lambda)] + \eta(p - \tau_n)[1 - \beta(1 - \lambda - \alpha_{wn})]}{1 - \beta(1 - \lambda - \eta\alpha_{wn})}. \quad (\text{A.5})$$

Expression (A.5) implies $\partial w_n / \partial \tau_n < 0$: higher non-green taxes depress non-green wages, holding tightness (hence α_{wn}) constant. $\square$

C.2 Green Jobs

Proof. The green sector value functions (see equations (16), (18), and (11) in the main text) are:

$$U_g = z - (1 - s_g)\kappa_g + \beta[\alpha_{wg}W_g + (1 - \alpha_{wg})U_g], \quad W_g = w_g + \beta[(1 - \lambda)W_g + \lambda U_g],$$

$$J_g = (p + \tau_g - w_g) + \beta[(1 - \lambda)J_g + \lambda V],$$

with $V = 0$ by free entry.

Step 1: Solving for U_g . From the law of motion for U_g ,

$$U_g = z - (1 - s_g)\kappa_g + \beta\alpha_{wg}W_g + \beta(1 - \alpha_{wg})U_g$$

so that

$$U_g(1 - \beta(1 - \alpha_{wg})) = z - (1 - s_g)\kappa_g + \beta\alpha_{wg}W_g,$$

and hence

$$U_g = \frac{z - (1 - s_g)\kappa_g}{1 - \beta(1 - \alpha_{wg})} + \frac{\beta\alpha_{wg}}{1 - \beta(1 - \alpha_{wg})} W_g. \quad (\text{A.6})$$

Step 2: Solving for W_g . From the law of motion for W_g ,

$$W_g = w_g + \beta(1 - \lambda)W_g + \beta\lambda U_g \implies W_g(1 - \beta(1 - \lambda)) = w_g + \beta\lambda U_g.$$

Substitute (A.6):

$$W_g(1 - \beta(1 - \lambda)) = w_g + \beta\lambda \left[\frac{z - (1 - s_g)\kappa_g}{1 - \beta(1 - \alpha_{wg})} + \frac{\beta\alpha_{wg}}{1 - \beta(1 - \alpha_{wg})} W_g \right].$$

Rearranging,

$$W_g \left(1 - \beta(1 - \lambda) - \frac{\beta^2\lambda\alpha_{wg}}{1 - \beta(1 - \alpha_{wg})} \right) = w_g + \frac{\beta\lambda}{1 - \beta(1 - \alpha_{wg})} (z - (1 - s_g)\kappa_g).$$

Combine terms as before:

$$W_g \frac{(1 - \beta)(1 - \beta(1 - \alpha_{wg} - \lambda))}{1 - \beta(1 - \alpha_{wg})} = w_g + \frac{\beta\lambda}{1 - \beta(1 - \alpha_{wg})} (z - (1 - s_g)\kappa_g).$$

Therefore,

$$W_g = \frac{w_g(1 - \beta(1 - \alpha_{wg})) + \beta\lambda(z - (1 - s_g)\kappa_g)}{(1 - \beta)(1 - \beta(1 - \alpha_{wg} - \lambda))}. \quad (\text{A.7})$$

Step 3: Substituting back into U_g . Using (A.6) and (A.7),

$$U_g = \frac{z - (1 - s_g)\kappa_g}{1 - \beta(1 - \alpha_{wg})} + \frac{\beta\alpha_{wg}}{1 - \beta(1 - \alpha_{wg})} \cdot \frac{w_g(1 - \beta(1 - \alpha_{wg})) + \beta\lambda(z - (1 - s_g)\kappa_g)}{(1 - \beta)(1 - \beta(1 - \alpha_{wg} - \lambda))}.$$

Simplifying,

$$U_g = \frac{(1 - \beta(1 - \lambda))(z - (1 - s_g)\kappa_g) + \beta\alpha_{wg}w_g}{(1 - \beta)(1 - \beta(1 - \alpha_{wg} - \lambda))}. \quad (\text{A.8})$$

Step 4: Surplus on the worker side. Subtract (A.8) from (A.7):

$$W_g - U_g = \frac{w_g - (z - (1 - s_g)\kappa_g)}{1 - \beta(1 - \alpha_{wg} - \lambda)}.$$

Step 5: Firm surplus and Nash bargaining. With $V = 0$, the job value satisfies

$$J_g = (p + \tau_g - w_g) + \beta(1 - \lambda)J_g \implies J_g = \frac{p + \tau_g - w_g}{1 - \beta(1 - \lambda)}.$$

The Nash condition in the green sector is

$$(1 - \eta)(W_g - U_g) = \eta J_g.$$

Substitute the surplus terms:

$$(1 - \eta) \frac{w_g - (z - (1 - s_g)\kappa_g)}{1 - \beta(1 - \alpha_{wg} - \lambda)} = \eta \frac{p + \tau_g - w_g}{1 - \beta(1 - \lambda)}.$$

Solve for w_g :

$$w_g \left[\frac{1 - \eta}{1 - \beta(1 - \alpha_{wg} - \lambda)} + \frac{\eta}{1 - \beta(1 - \lambda)} \right] = \frac{(1 - \eta)(z - (1 - s_g)\kappa_g)}{1 - \beta(1 - \alpha_{wg} - \lambda)} + \frac{\eta(p + \tau_g)}{1 - \beta(1 - \lambda)}.$$

Collecting terms over the common denominator $1 - \beta(1 - \lambda - \eta\alpha_{wg})$, we obtain

$$w_g = \frac{(1 - \eta)(z - (1 - s_g)\kappa_g) [1 - \beta(1 - \lambda)] + \eta(p + \tau_g) [1 - \beta(1 - \lambda - \alpha_{wg})]}{1 - \beta(1 - \lambda - \eta\alpha_{wg})}. \quad (\text{A.9})$$

Expression (A.9) implies $\partial w_g / \partial \tau_g > 0$, $\partial w_g / \partial s_g > 0$, and $\partial w_g / \partial \kappa_g < 0$, holding tightness (hence α_{wg}) constant. $\square$

Equations (A.5) and (A.9) summarize the wage schedules.